

12.2 Example Reports

To ensure clarity and consistency in reporting, it is recommended to establish a standardized format that aligns to ISO and specific customer requirements. The following key elements are expected to be included in a process capability report. This section provides examples of reporting formats for both normal and non-normal distribution process capability studies:

- **Figure 12-1:** Example Report for Process Capability (Normal Distribution)
- **Figure 12-2:** Example Report for Process Capability (Non-normal Distribution)
- **Figure 12-3:** Example Report for Process Capability (Non-normal Distribution)

Note: The reporting format and content included must be mutually agreed upon by both the customer and the supplier.

List of possible reporting elements

1. Name of Process, Machine and location study was performed
2. Reference numbers associated with Process and Machine
3. People who performed the study and took the measurements
4. Time and duration of the data collection
5. Name and reference number for the part
6. Identified part characteristic & Specification Limits (units of measure)
7. Technical conditions under which the study was run (batches, operation, tools)
8. Deviations from defined stable operating conditions (environment, process settings, cycle time)
9. Number of values used for calculation
10. Frequency of sampling & subgroup size
11. Histogram of the data
12. A Run chart for qualitative assessment of data over time
13. Probability plot of the data
14. Control Chart for stability assessment
15. Estimated parameter for process location with calculation method
16. Estimated parameter for process variation with calculation method
17. Selected distribution model used for estimates
18. Performance / capability requirements / Calculation method
19. Calculated performance / capability indices, estimated % out of specification (with confidence intervals as required)

20. Conclusions & Recommendation, Corrective actions

Other elements that may be included:

21. Time dependent distribution model

22. Measurement uncertainty and guard bands for 100% inspections or conformity testing in case of out-of-control or out-of-specification situations.

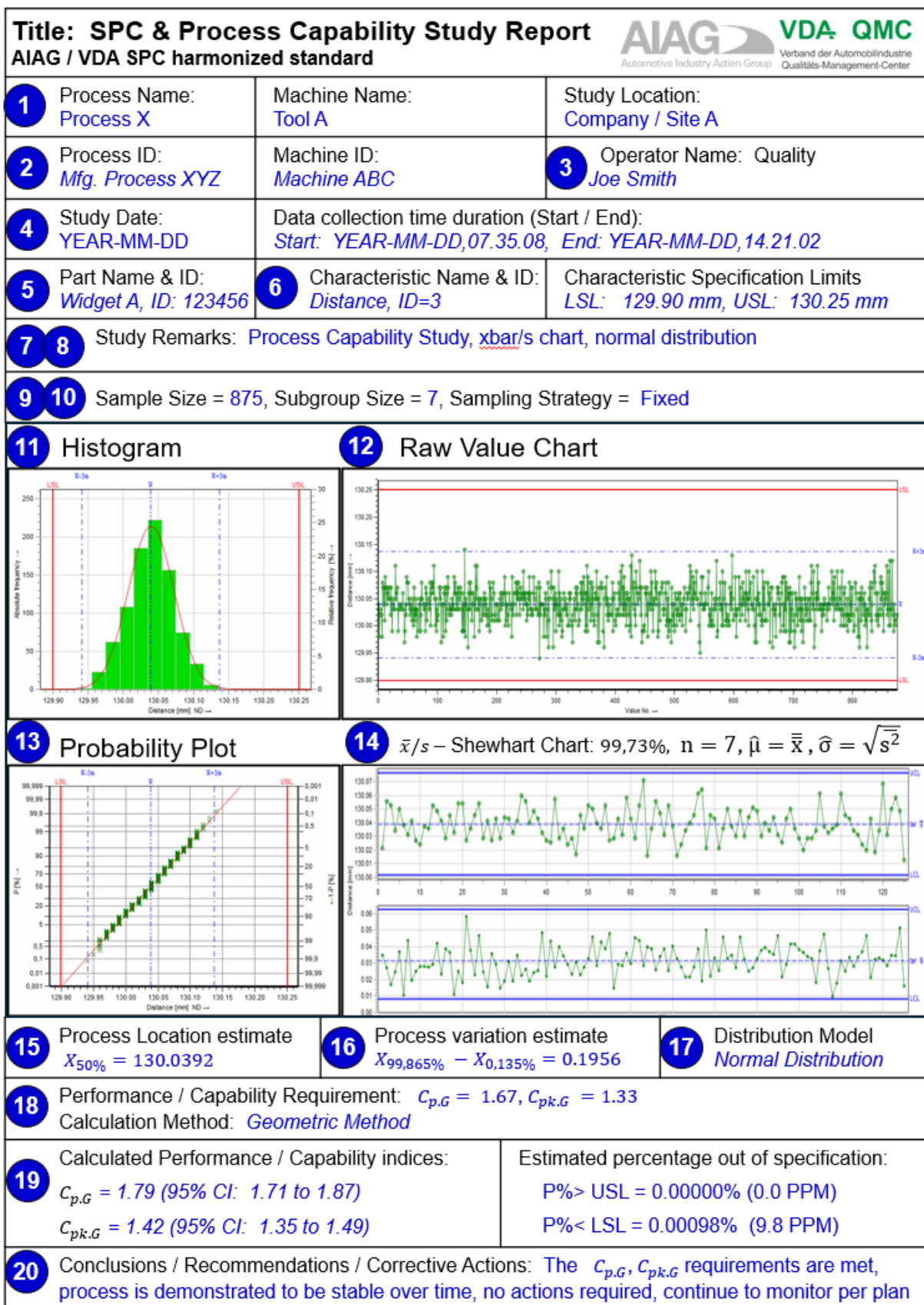


Figure 12-1: Example Report with Fictional Data for Process Capability (Normal Distribution)

Summary of Key Report Observations, as illustrated in Figure 12-1

- **Element 11:** Histogram indicates the data appears to be normally distributed but not centered within specification limits
- **Element 12:** Process location and variation appear to be consistent over time, no outliers, shifts or drifts observed
- **Element 13, 17:** Probability plot confirms the distribution selection, “Normal distribution” to be used for performance/capability estimates
- **Element 14:** Process is in a state of statistical control, stable over time (no points outside of control limits)
- **Element 18,19:** Performance/capability indices are calculated by the Geometric method, estimates are $C_{p.G} = 1.79$, $C_{pk.G} = 1.42$
- **Element 20:** Both $C_{p.G}$ & $C_{pk.G}$ indices meet the targets requirements, even at the lower bound of 95% confidence interval. Capability indices labeled “C” are used because the process study was sufficient to confirm stability over time, capturing the full scope of process variation across all 5M factors.

Title: SPC & Process Capability Study Report

AIAG / VDA SPC harmonized standard



1 Process Name: <i>Process X</i>	Machine Name: <i>Tool A</i>	Study Location: <i>Company / Site A</i>
2 Process ID: <i>Mfg. Process XYZ</i>	Machine ID: <i>Machine ABC</i>	3 Operator Name: <i>Quality Joe Smith</i>
4 Study Date: <i>YEAR-MM-DD</i>	Data collection time duration (Start / End): <i>Start: YYYY-MM-DD, hh:mm:ss, End: YYYY-MM-DD, hh:mm:ss</i>	
5 Part Name & ID: <i>Widget A, ID: 123456</i>	6 Characteristic Name & ID: <i>Distance, ID=3</i>	Characteristic Specification Limits <i>LSL: 29.860 mm, USL: 0.280 mm</i>
7 8 Study Remarks: <i>Process Capability Study with fixed tooling change</i>		
9 10 Sample Size = <i>500</i> , Subgroup Size = <i>5</i> , Sampling Strategy = <i>Fixed</i>		
11 Histogram		12 Raw Value Chart
13 Probability Plot	14 $\bar{x}/s$ - Extended Shewhart Chart: 99% [$n = 5, \hat{\mu} = \bar{\bar{x}}, \hat{\sigma} = \sqrt{s^2}$]	
15 Process Location estimate $X_{50\%} = 30.0092$	16 Process variation estimate $X_{99,865\%} - X_{0,135\%} = 0,16188$	17 Distribution Model <i>Mixed Distribution</i>
18 Performance / Capability Requirement: $C_{p,G} = 1.67, C_{pk,G} = 1.33$ Calculation Method: <i>Geometric Method</i>		
21 Time Dependent Distribution Model: <i>C4</i>	22 Exp. Measurement Uncertainty: <i>0.0211</i> Guard Band (95%): <i>0.0174</i>	
19 Calculated Performance / Capability indices: $C_{p,G} = 1.73$ (95% CI: 1.62 to 1.84) $C_{pk,G} = 1.67$ (95% CI: 1.57 to 1.78)		Estimated percentage out of specification: $P\% > USL = 0.00000\%$ (0.0 PPM) $P\% < LSL = 0.00000\%$ (0.0 PPM)
20 Conclusions / Recommendations / Corrective Actions: <i>The $C_{p,G}, C_{pk,G}$ requirements are met, process is demonstrated to in control, no actions required, continue to monitor per plan</i>		

Figure 12-2: Example Report with Fictional Data for Process Capability (Non-Normal Distribution)

Interpretation of the Report Elements

What can be identified in the sample report? :

1. The process location is not constant over time (12) due to fixed tooling and tool change (7)(8).
2. Variation seems to be constant over time (14).
3. There seem to be no outliers (12).
4. The process is not normally distributed (11).
5. A multimode distribution is appropriate (13), and a mixed distribution based on the EM-Method (Expectation-Maximization-Method) with up to 5 kernel is used (11)(17).
6. The assumed distribution is confirmed in the probability plot (13).
7. Therefore, we assume distribution time model C4 (21).
8. The calculated performance/capability is $C_{p.G} = 1.73$ and $C_{pk.G} = 1.67$ (19), calculated with general geometric (quantile) method (18).
9. Both indices exceed the target indices (18), $C_{pk.G}$ even with a lower confidence limit (19) based on a 95% confidence interval.
10. The process shows no out-of-control situations in the control chart (14). The control chart is in this case an Extended Shewhart Chart.
11. Since the process is in control (14), the resulting indices are named C instead of P (19).
12. In case of an unexpected future out-of-control situation, the expanded uncertainty of the measurement system is 0.0211 mm, and the guard band is 0.0175 mm for 5% error probability (22).

Another example of a report of the same dataset as example 2 with additional information is shown in Figure 12-3. The table allows a direct comparison of the drawing data with the measured values and statistical results.:

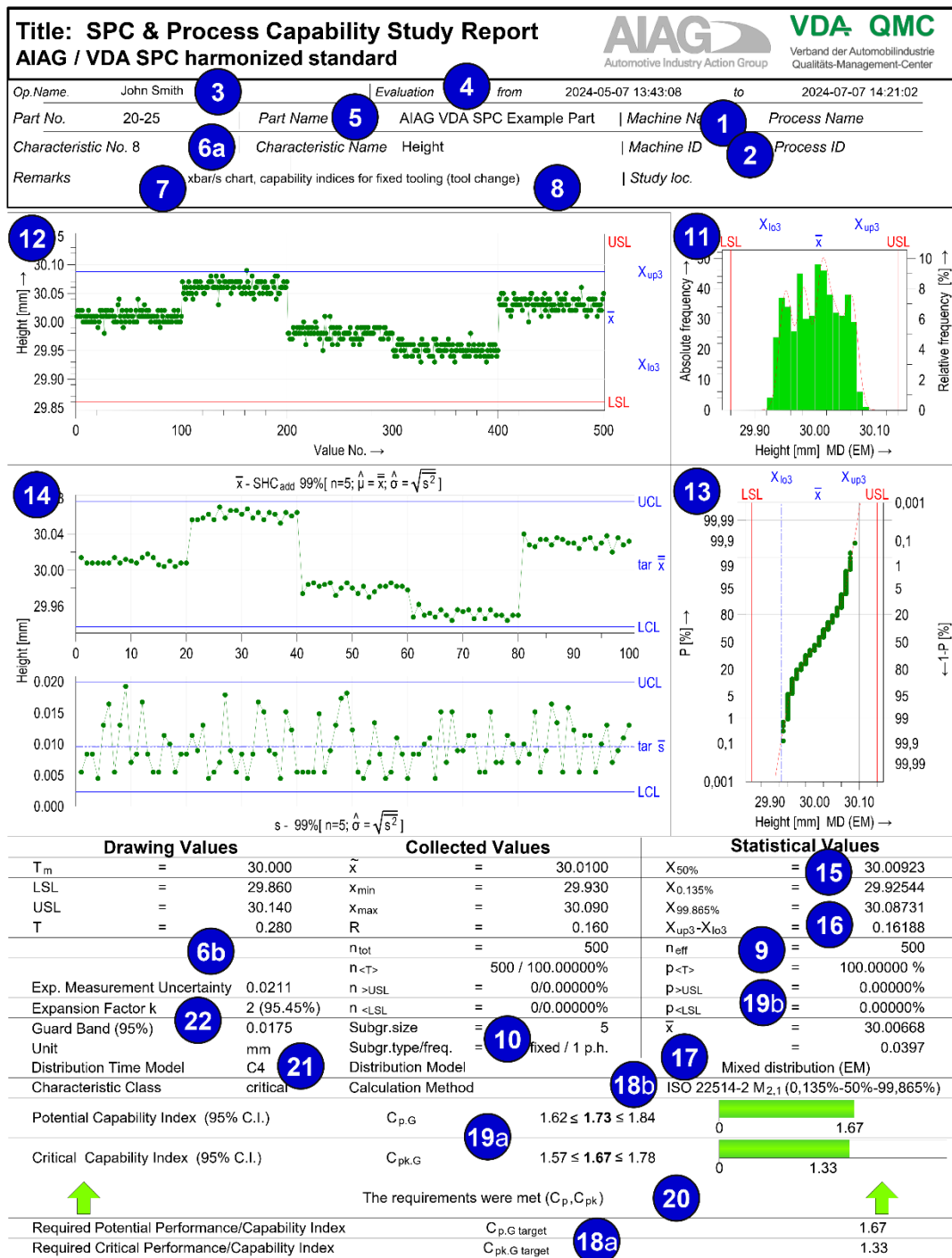


Figure 12-3 Example Report with Fictional Data for Process Capability (Non-Normal Distribution)